SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1005	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	22.10.2026	**Duration:**	60 minutes

Code of Conduct

*I , **Dharshavardhan 192420039** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement**
   - Explain the problem in your own words.
   - Identify the objectives of the proposed system.
   - State the scope of the problem.
2. **Identify Key Stakeholders and Challenges**
   - Identify all the stakeholders involved in the system.
   - Explain the roles and responsibilities of each stakeholder.
   - Discuss the major challenges and issues faced by the stakeholders.
3. **Analyze the Current Approach**
   - Describe the existing system or current process.
   - Identify its strengths and limitations.
   - Analyze the problems or gaps that need to be addressed.

4. **Propose a Suitable Solution Using Software Engineering Principles**

    - Design a software-based solution for the given problem.
    - Describe the major functional modules of the proposed system.
    - Identify the functional and non-functional requirements.
    - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. **Justify the Chosen Methodology/Framework**

    - Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
    - Justify why the selected methodology/framework is suitable for the proposed system.
    - Explain how it supports successful project development and maintenance.

6. **Discuss Expected Outcomes and Limitations**

    - Describe the expected benefits of the proposed solution.
    - Explain how the solution addresses the identified challenges.
    - Discuss potential risks, implementation challenges, and limitations.
    - Suggest possible future enhanceme

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification	Selects the most suitable SDLC/model/fram	Suitable methodology selected with	Methodology selected but justification is	Outcomes, limitations, or future enhancements

(10%)	ework and provides strong technical justification.	adequate justification.	weak.	are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

1. Understand the Problem Statement

1.1 Problem Explanation

Academic conferences receive research papers from many authors and must coordinate submission, reviewer selection, evaluation, revisions, final decisions, and camera-ready paper collection. When these activities are handled through emails, spreadsheets, or separate tools, administrators may face duplicated records, missed deadlines, reviewer conflicts, delayed decisions, and difficulty tracking each paper. The proposed Smart Conference Paper Submission and Review System provides a centralized web-based platform to manage the complete paper-review lifecycle.

1.2 Objectives

- Manage author and reviewer registrations and user roles.
- Accept research paper submissions and maintain submission records.
- Allocate papers to suitable reviewers while considering expertise and conflicts of interest.
- Collect reviewer comments, scores, and recommendations confidentially.
- Track review deadlines, author revisions, and camera-ready submissions.
- Support acceptance, rejection, and revision decisions with notifications.
- Generate reports on submissions, review progress, decisions, and conference statistics.

1.3 Scope

The system covers registration, paper upload, reviewer assignment, conflict-of-interest declarations, review submission, revision management, decision notification, final-paper collection, deadline reminders, and conference analytics. It is intended for conference chairs, administrators, authors, and reviewers. It does not itself guarantee the scientific quality of a paper or replace the academic judgment of reviewers and program committee members.

2. Identify Key Stakeholders and Challenges

2.1 Stakeholders and Responsibilities

Stakeholder	Responsibilities	Main Challenges
Authors	Register, submit papers, respond to reviews, upload revisions and final papers.	Submission errors, unclear status, and missed deadlines.
Reviewers	Declare conflicts, review assigned papers, provide scores and comments.	Workload, limited time, confidentiality, and conflict management.
Conference chair / program committee	Set policies, oversee assignments, resolve disputes, and make final decisions.	Fair allocation, timely decisions, and consistent review standards.
Conference administrator	Configure conference dates, manage accounts, monitor progress, and issue reports.	Large volumes of records and repetitive administrative work.
System administrator	Maintain accounts, access control, backups, availability, and security.	System reliability, data protection, and technical support.

2.2 Major Challenges

- Managing a large number of submissions and revisions without losing records.
- Matching papers to reviewers based on subject expertise and workload.
- Preventing reviewers from assessing papers where a conflict of interest exists.
- Maintaining reviewer confidentiality and protecting unpublished research.
- Tracking different deadlines and ensuring that notifications are delivered on time.
- Maintaining consistent records of reviews, decisions, and final versions.

3. Analyze the Current Approach

3.1 Existing Process

In a manual or partially digital process, authors may submit papers by email or through basic forms. Administrators maintain spreadsheets, assign reviewers manually, collect feedback through

email, and communicate decisions individually. Revised and final files may be stored in separate folders.

3.2 Strengths

- Low initial setup cost for small conferences.
- Familiar tools such as email and spreadsheets can be used without extensive training.
- Flexible for small events with a limited number of papers.

3.3 Limitations and Gaps

- Records can become duplicated, incomplete, or inconsistent.
- Manual reviewer allocation can create workload imbalance or conflicts of interest.
- Email-based reviews make it difficult to track status and protect confidentiality.
- Deadlines and revision versions may be overlooked.
- Generating accurate conference statistics requires additional manual work.
- Access permissions may not reliably separate author, reviewer, and chair information.

4. Proposed Solution Using Software Engineering Principles

4.1 Proposed System

The proposed solution is a role-based web application with a central database and secure file storage. Users sign in according to their role. Authors submit papers and revisions; the system validates required information and records version history. Administrators or program chairs allocate reviewers using topic expertise, availability, workload, and declared conflicts. Reviewers submit confidential evaluations. Chairs review recommendations and record decisions. The system notifies authors and tracks camera-ready files and deadlines.

4.2 Major Functional Modules

- User Registration and Authentication: account creation, login, role management, and profile updates.
- Conference and Submission Management: conference details, paper metadata, file upload, validation, and submission tracking.
- Reviewer Allocation and Conflict Management: expertise-based assignment, workload checks, conflict declarations, and reassignment.

- Review Management: review forms, scores, comments, reviewer deadlines, and confidential access.
- Decision and Revision Management: chair decisions, author revision uploads, version tracking, and status updates.
- Camera-Ready Collection: final manuscript upload, required-file checks, and completion status.
- Notification and Deadline Tracking: reminders for submissions, reviews, revisions, and final-paper deadlines.
- Analytics and Reporting: totals for submissions, pending reviews, acceptance/rejection decisions, and completion rates.

4.3 Proposed System Architecture

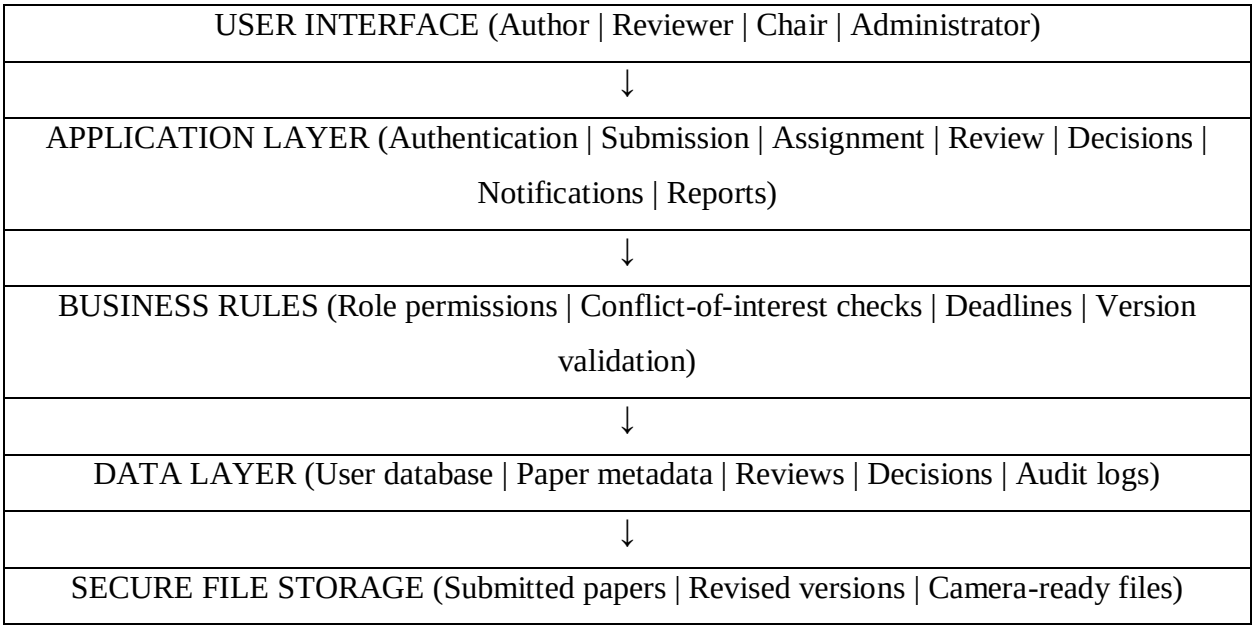


Figure 1. High-level architecture of the proposed system. The layers separate the user interface, application functions, business rules, stored records, and uploaded files.

4.4 Typical Workflow

1. Administrator creates a conference and sets submission, review, revision, and final-paper deadlines.
2. Author registers, enters paper details, and uploads a manuscript.
3. System validates required fields and stores the submission securely.
4. Chair or administrator assigns reviewers based on expertise, workload, and conflict declarations.

5. Reviewers submit evaluations and recommendations before the deadline.
6. Chair considers reviews and records the decision.
7. Author receives the decision and, when requested, uploads a revised paper.
8. Accepted authors upload the camera-ready version; the administrator verifies completion.
9. System updates dashboards and generates conference reports.

4.5 Functional Requirements

- The system shall allow users to register, log in, and access role-specific functions.
- The system shall accept paper metadata and manuscript files and provide a submission reference.
- The system shall allow authorized staff to assign and reassign reviewers.
- The system shall record conflict-of-interest declarations and prevent conflicted assignments where declared.
- The system shall allow reviewers to submit scores, comments, and recommendations securely.
- The system shall support decisions, revisions, version tracking, and camera-ready uploads.
- The system shall send deadline and decision notifications.
- The system shall generate reports on submission counts, review status, and decisions.

4.6 Non-Functional Requirements

- Security: authenticated access, role-based permissions, secure file handling, and protected review data.
- Confidentiality: reviewer identities and confidential comments are visible only to authorized roles according to conference policy.
- Reliability: regular backups, error handling, and recovery procedures.
- Usability: clear dashboards, accessible forms, and understandable status messages.
- Performance: common operations should remain responsive during submission peaks.
- Scalability: support additional papers, reviewers, and conferences with minimal redesign.
- Maintainability: modular code, documented interfaces, and testable components.
- Auditability: record important actions such as assignments, decisions, and file-version changes.

4.7 Software Engineering Principles

Modularity: Separate registration, submission, review, decision, notification, and reporting functions so each can be developed and tested independently.

Scalability: Use a structured database and deployable application services that can be expanded as the conference grows.

Maintainability: Use clear naming, documentation, version control, and automated tests to simplify updates.

Reliability: Validate inputs, handle errors, back up records, and maintain traceable paper versions.

Usability: Provide role-specific dashboards, simple forms, and visible deadline/status indicators.

Security: Apply least-privilege permissions, secure authentication, protected storage, and audit logging.

5. Justify the Chosen Methodology / Framework

5.1 Selected Methodology: Agile Scrum

Agile Scrum is suitable because the system contains several independent modules and conference requirements may change after feedback from chairs, reviewers, and authors. Scrum supports development in short iterations, regular demonstrations, prioritized backlogs, and continuous testing. A small team can deliver a working core system first and add advanced functions in later sprints.

5.2 Suggested Sprint Plan

Sprint	Main Work	Deliverable
Sprint 1	Requirements, user roles, UI prototype, and database design.	Approved requirements and prototype.
Sprint 2	Registration, login, conference setup, and paper submission.	Working submission module.

Sprint 3	Reviewer profiles, allocation, conflict declarations, and review forms.	Working reviewer workflow.
Sprint 4	Decision, revision, camera-ready collection, and notifications.	End-to-end review workflow.
Sprint 5	Analytics, security testing, usability testing, bug fixing, and deployment.	Tested release and documentation.

5.3 Support for Development and Maintenance

At the end of each sprint, stakeholders can review the working increment and suggest changes. Continuous integration and testing help detect defects early. A prioritized backlog allows urgent security or workflow issues to be addressed first. After deployment, feedback and defect reports can be converted into maintenance tasks.

6. Expected Outcomes and Limitations

6.1 Expected Outcomes

- Centralized and searchable conference records.
- Reduced repetitive administrative work and fewer manual tracking errors.
- More organized reviewer allocation and improved conflict handling.
- Clear tracking of reviews, revisions, decisions, and final submissions.
- Improved confidentiality through role-based access control.
- Timely reminders and more consistent communication.
- Faster generation of conference progress and decision reports.

6.2 Risks and Implementation Challenges

- Incorrect or incomplete conflict declarations may still result in unsuitable assignments.
- Users may require training or support when moving from email-based workflows.
- Large file uploads or peak submission periods may affect performance if capacity is insufficient.
- Notification delivery can fail because of invalid addresses or external email service issues.

- Sensitive unpublished papers and reviews require careful security, backup, and access-control practices.
- Requirements and review policies differ between conferences and may require configuration.

6.3 Limitations

The system can assist with workflow and consistency, but it cannot guarantee reviewer fairness, eliminate all conflicts, or determine scientific merit automatically. Its effectiveness depends on accurate user information, clear conference policies, secure infrastructure, and timely participation by reviewers and authors.

6.4 Future Enhancements

- Integrate ORCID or institutional single sign-on for identity verification.
- Add configurable double-blind or single-blind review policies.
- Provide reviewer expertise suggestions using paper keywords and reviewer profiles, with human approval.
- Integrate plagiarism-screening services where licensed and permitted.
- Support conference proceedings export and DOI metadata preparation.
- Add multilingual interfaces, accessibility improvements, and mobile-friendly dashboards.

7. Conclusion

The Smart Conference Paper Submission and Review System offers a structured way to manage the complete conference paper lifecycle, from registration and submission to review, decisions, revisions, and final-paper collection. By combining role-based access, conflict-of-interest handling, deadline tracking, notifications, and analytics, it addresses major weaknesses of manual processes. An Agile Scrum approach enables the system to be developed incrementally, tested continuously, and refined through stakeholder feedback.

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